

Practice III: Measuring Disciplinary Hardness and Softness from Scientific Networks

Objective

In Practices I and II, participants:

1. Extracted records from Web of Science.
2. Built coreference and coauthorship networks.

In **Practice III**, you will use these networks to measure **disciplinary hardness and softness**. The goal is to understand how network structure reflects differences between scientific fields and research traditions.

Network Indicators

1. **Average Clustering Coefficient.** The clustering coefficient measures local cohesion.
For node i :

$$C_i = \frac{2e_i}{k_i(k_i - 1)}$$

where:

- k_i = number of neighbors
- e_i = number of ties among those neighbors

We compute: $\bar{C}$

the average clustering coefficient across the network.

2. **Average Degree Centrality.** Degree centrality measures the proportion of nodes to which a node is directly connected.

$$C_D(i) = \frac{k_i}{n - 1}$$

where:

- k_i = degree of node i
- n = number of nodes

We compute: $\bar{C}_D$

the average degree centrality across all nodes.

- 3. Average Closeness Centrality.** Closeness centrality measures how quickly a node can reach all other nodes.

$$C_c(i) = \frac{n-1}{\sum_j d(i,j)}$$

where $d(i,j)$ is the shortest path distance. We compute: $\bar{C}_C$
the average closeness centrality across the network.

4. Average Betweenness Centrality

Betweenness centrality measures how frequently a node lies on shortest paths connecting other nodes.

$$C_B(i)$$

We compute:

$$\bar{C}_B$$

the average betweenness centrality across all nodes.

Exercise

Using the networks generated in Practice II:

Calculate:

- Average Clustering Coefficient
- Average Degree Centrality
- Average Closeness Centrality
- Average Betweenness Centrality

for both:

- Coreference Network
- Coauthorship Network

Results Table

Create the following table:

Network	Avg Degree	Avg Closeness	Avg Betweenness	Avg Clustering	Seed Clustering
Coreference					
Coauthorship					